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Reacting Radiating Fuel Ring**

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Grid-Free Simulation of a Reacting Radiating Fuel Ring

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Simulations of the combustion of an axisymmetric fuel ring in which thermal radiation is incorporated using a novel grid-free method, and diffusive transport of vorticity and scalars are implemented utilizing accurate Lagrangian algorithms are presented. Radiative transport is simulated using the discrete-source method in which the irradiance of a finite-core hot gas element is obtained by approximating the volumetric integral around the center of the element while using an effective attenuation coefficient between the source and the target. Diffusion is simulated using a smooth redistribution method in which the same core function is used to discretize and reconstruct the diffusing variables. Convergence studies are presented. Results show that vorticity generation due to buoyancy and baroclinicity imparts a motion on the ring, which, at the early stages, maintains its initial symmetry, while at the later stages, causes a noticeable distortion of the ring structure. The later is associated with the dissipation of vorticity across the axis of symmetry. Radiation reduces the total thermal energy retained by the ring, and redistributes this fraction so as to reduce the temperature gradients inside, thus by weakening the vorticity generation mechanisms.

Nomenclature

a	Absorption coefficient
c_p	Specific heat at constant pressure
$\mathcal{D}$	Mass diffusion coefficient
g	Gravity
M_i	Molecular mass of species i
p	Thermodynamic pressure
Q_f	Heat of reaction of fuel
$\mathbf{q}_r$	Radiative heat flux
$\mathbf{r}$	Axisymmetric coordinates; $\mathbf{r} \equiv r\hat{\mathbf{r}} + z\hat{\mathbf{z}}$
s	Shvab-Zel'dovich scalar
T	Temperature
$\mathbf{u}$	Velocity vector
U	Reference velocity
Y_i	Mass fraction of species i
Γ	Circulation
λ	Thermal conductivity
μ	Coefficient of viscosity
ω	Vorticity
ϕ	Fuel to air mass stoichiometric ratio
ρ	Density
ϖ	Reaction rate

Subscripts

0	Freestream conditions
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Operators

$D\mathbf{u}/Dt$	Convective derivative
∇	Gradient operator
$\nabla \cdot \mathbf{u}$	Divergence of vector $\mathbf{u}$

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Introduction

VORTEX methods have been successful in simulating flows in which vorticity is localized in small regions of the domain. Examples include vortex sheets,¹⁻³ high Reynolds number wakes,^{4,5} three dimensional problems,⁶⁻⁹ reacting flows in shear layers,^{10,11} co-axial jets,¹² and isothermal and fire plumes.¹³⁻¹⁵ The grid-free nature of vortex methods is suitable for flows within complex geometries. The Lagrangian nature avoids discretization of the convective terms and the concomitant introduction of numerical diffusion; a problem that might destroy small scale features associated with slightly viscous flows. Additional advantages are the intrinsic satisfaction of infinity boundary conditions and the adaptive representation of the vorticity field in time as the elements move with fluid particles. An overview of vortex methods can be found in references 16-18.¹⁶⁻¹⁸

Some of the challenges facing vortex methods are accurate modeling of anisotropic diffusion, flows with large heat release, radiation, and handling of source terms in the vorticity, energy, and species conservation equations. The example presented in this work is based on recent developments, e.g., accurate modeling of diffusion via the smoothed redistribution method,¹⁹ and grid-free modeling of radiation via the discrete source method.²⁰

In combustion systems, radiation may lead to a substantial amount of energy loss. In fires, for example, the ratio of the total energy lost by radiation to the total heat released varies from 0.25 for methane flames^{21,22} to as large as 0.5 for acetylene flames. These experimental observations reveal the significance of radiation in studying combustion dy-

namics and its impact on its surroundings.

Radiative flux depends non-linearly on the instantaneous composition of the combustion products, their optical properties, and the temperature. Therefore, prediction of the radiative flux within the reaction zone and to surrounding objects requires accurate knowledge of distribution of these quantities.

The objective of this work is to present an application based on the progress reported above. The example is that of the burning of a fuel patch (ring) in an unbounded axisymmetric domain.

Governing Equations

The equations^{23,24} governing conservation of mass, momentum, energy, and chemical species for a multi-component reacting mixture of ideal gases are presented in dimensionless form as follows:

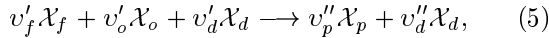
$$\frac{D\rho}{Dt} + \rho \nabla \cdot \mathbf{u} = 0, \quad (1)$$

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \frac{1}{Re_0} \mu \nabla^2 \mathbf{u} + (1 - \rho) \dot{\mathbf{z}}, \quad (2)$$

$$\rho c_p \frac{DT}{Dt} = \frac{1}{Pe_0} \nabla \cdot (\lambda \nabla T) + Q_f \varpi_f - \nabla \cdot \mathbf{q}_r, \quad (3)$$

$$\rho \frac{DY_i}{Dt} = \frac{1}{Le_0 Pe_0} \nabla \cdot (\rho \mathcal{D} \nabla Y_i) + \Lambda_i \varpi_f, \quad i = 1, \dots, N_s. \quad (4)$$

The chemical reaction, assumed single-step and irreversible, accounts for consumption of reactants, fuel (f) and oxidant (o), in the presence of a diluent (d), and production of products (p) according to



where v'_i and v''_i are the stoichiometric coefficients for species i , appearing as a reactant and as a product, respectively; $i = f, o, d, p$. The stoichiometric coefficients specify the molecular proportions in which the reactants participate. The chemical symbol for species i is denoted by $\mathcal{X}_i$.

Ideal Gas equation of state:

$$\frac{\rho T}{M} = 1. \quad (6)$$

In the energy equation, Eq. (3), $\Lambda_f = -1$, $\Lambda_o = -(M_o v'_o)/(M_f v'_f) \equiv -\phi$, $\Lambda_p = (M_p v'_p)/(M_f v'_f) \equiv 1 + \phi$, and $\Lambda_d = 0$. The Reynolds, Prandtl, Peclet, and Lewis numbers are respectively

$$Re_0 = \frac{U r_0}{\nu_0}, \quad Pr_0 = \frac{\nu_0}{\alpha_0}, \quad Pe_0 = \frac{r_0 U}{\alpha_0}, \quad Le_0 = \frac{\alpha_0}{D_0}, \quad (7)$$

where the thermal and kinematic diffusivities are $\alpha_0 = \lambda_0/\rho_0 c_{p0}$ and $\nu_0 = \mu_0/\rho_0$. For nondimensionalization, the reference parameters used are initial radius, r_0 , buoyancy characteristic velocity $U \equiv \sqrt{g r_0}$, and properties of air at atmospheric conditions.

Numerical Methodology

The numerical methodology is based on the vortex method in which the vorticity field is discretized among computational elements which move along particles trajectories. The vorticity field is represented¹⁶ according to

$$\omega(\mathbf{r}, t) = \sum_{i=1}^N \Gamma_i(t) \phi_\sigma(\mathbf{r}; \mathbf{r}_i), \quad (8)$$

where $\Gamma_i(t)$ is the circulation of element i and ϕ is the core function with core radius σ . Vorticity is transported in time according to the vorticity equation

$$\frac{D\omega}{Dt} = \left(\frac{u_r}{r} - \nabla \cdot \mathbf{u} \right) \omega + \frac{1}{\rho} \left(a_z \frac{\partial \rho}{\partial r} - a_r \frac{\partial \rho}{\partial z} \right) + \frac{\nu}{Re_0} \tilde{\nabla}^2 \omega, \quad (9)$$

where $a_r = -Du_r/Dt$, $a_z = -1 - Du_z/Dt$, and $\tilde{\nabla}^2 \equiv \nabla^2 - 1/r^2$.

Eq. (9) is solved using Strang viscous splitting²⁵ into an inviscid component and a viscous component. Noting that circulation of a material element is conserved along a particle path for an inviscid flow,^{24,26} this component reduces to

$$\frac{d\Gamma_i}{dt} = 0. \quad (10)$$

The viscous component, governed by $\partial\omega/\partial t = \tilde{\nabla}^2 \omega$, is solved using the smoothed redistribution method,¹⁹ where diffusion of an element is simulated by transferring fractions of its circulation to neighboring elements while adjusting its core radius. The smoothed redistribution extends the (point) redistribution method⁴ by incorporating a smooth core function (as opposed to a Dirac delta function) thus allowing for reconstruction of smooth and accurate vorticity field according to Eq. (8). The core function utilized, $\phi_\sigma(\mathbf{r}; \mathbf{r}_i)$, is based on the Green's function of the diffusion equation.

The vorticity generation (except for decay due to diffusion across the axis of symmetry) component is solved by integrating the equation $d\omega/dt = 1/\rho (a_z \partial \rho / \partial r - a_r \partial \rho / \partial z)$ in time using a first order backward Euler scheme, yielding $\omega(\mathbf{r}, t + \Delta t)$. However, since the quantities transported are the elements' circulations, Eq. (8) has to be solved for $\Gamma_i(t + \Delta t)$. One way of doing this is to evaluate $\omega(\mathbf{r}, t + \Delta t)$ at element locations $\mathbf{r}_i$ which results in a linear system of size $N \times N$, N being the number of elements. Being an interpolation matrix, the coefficient matrix is ill conditioned. Further, exact solution of the linear system does not guarantee accurate and smooth solution everywhere, and a noisy oscillating solution may result. A remedy to this problem is proposed by Marshall and Grant.²⁷ Based on a Gauss-Seidel point iterative technique, the scheme assumes that the circulations of elements within a neighborhood radius of the current element are set temporarily to the current element's circulation, i.e.

$$\Delta \Gamma_i^{(m+1)} = \frac{\Delta t \Theta(\mathbf{r}_i) - \sum_{j \notin C(i)} \Delta \Gamma_j^{(m)} \phi_\sigma(\mathbf{r}_i, \mathbf{r}_j)}{\sum_{j \in C(i)} \phi_\sigma(\mathbf{r}_i, \mathbf{r}_j)}, \quad (11)$$

where m is the iteration level, C_i is the set of neighbors of current element i , and $\Theta(\mathbf{r})$ is the vorticity source term. The neighborhood radius is usually selected to be of the order of core radius σ . Iterations proceed until convergence.

Scalar Transport

Assuming that Lewis number for all species and Prandtl number are equal to unity, and the specific heat is constant, the energy and species conservation equations may be reduced to

$$\frac{Ds}{Dt} = \frac{1}{\rho Pe_0} \nabla^2 s - \frac{C_s}{\rho} \nabla \cdot \mathbf{q}_r. \quad (12)$$

Using Shvab-Ze'ldovich formulation,²³ s is $\eta_{sz} = Y_o - \phi Y_f$, or $\gamma_{sz} = T - Y_p Q_f / (1 + \phi)$, or $\delta_{sz} = Y_d$ for which $C_{\eta_{sz}} = 0$, $C_{\gamma_{sz}} = 1$, $C_{\delta_{sz}} = 0$ respectively.

The scalar field is discretized according to

$$s(\mathbf{r}, t) = \sum_{i=1}^N \frac{E_i(t)}{r_i} \phi_\sigma(\mathbf{r}; \mathbf{r}_i), \quad (13)$$

where the quantity transported is the energy $E_i(t)$ of element i and ϕ is the core function with core radius σ . The scalar equation, Eq. (12), is solved in a similar manner to the vorticity equation, as discussed in the previous subsection.

Radiation

We turn our attention now to the evaluation of the radiative source term in the energy equation. The medium is assumed to be gray and non-scattering. The thermal energy per unit volume absorbed by radiation^{28,29} at location $\mathbf{r}$ is then given by

$$-\nabla \cdot \mathbf{q}_r = -\frac{4aT^4}{Bo_0}(1 - G), \quad (14)$$

where a reference length r_0 is used to nondimensionalize a , the absorption coefficient. The Boltzmann number, evaluated for air at normal conditions as

$$Bo_0 = \frac{\rho_0 c_{p0} U}{\sigma^* T_0^3} = 4.1278113 \times 10^{-4} r_0^{-1/2}, \quad (15)$$

is the convection-to-radiation parameter, where σ^* is the Stefan-Boltzmann constant. The irradiance G , the amount of energy that reaches a certain location $\mathbf{r}$ nondimensionalized by the black body energy emitted at the same location, is

$$G(\mathbf{r}) = \frac{1}{4\pi i_b(\mathbf{r})} \int_V a(\mathbf{r}') i_b(\mathbf{r}') \frac{e^{-\int_{s'}^s a(s'') ds''}}{|\mathbf{r} - \mathbf{r}'|^2} dV(\mathbf{r}'), \quad (16)$$

where the volume V containing the radiating medium is selected large enough so that the radiation intensity crossing its boundary is negligible. The inner integral is along the path from the source point $\mathbf{r}'$ to the target point $\mathbf{r}$.

Calculating the irradiance numerically in a grid-free manner compatible with the vortex method is done

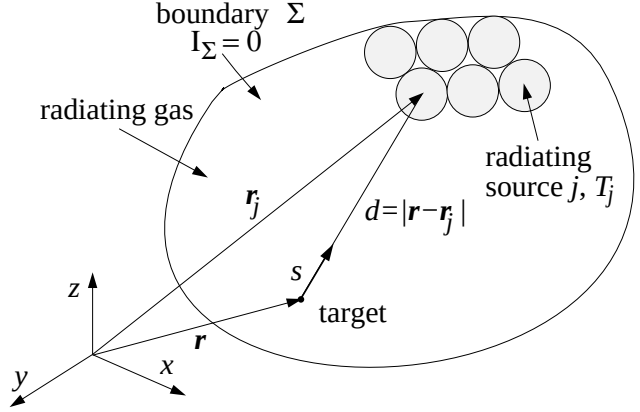


Fig. 1 Decomposing the participating medium into discrete source elements.

using the discrete source method.²⁰ Decomposing the volume V into N elemental volumes, and assuming that the temperature T_j and absorption coefficient a_j are uniform within each element, $j = 1, N$, then the irradiance may be expressed in the following discretized form,

$$G(\mathbf{r}) \simeq \sum_{j=1}^N G_j(\mathbf{r}), \quad (17)$$

where $G_j(\mathbf{r})$ is the irradiance induced by computational element j ,

$$G_j(\mathbf{r}) \simeq \frac{a_j i_{bj}}{4\pi i_b(\mathbf{r})} \int_{V_j} \frac{e^{-\bar{a}_j |\mathbf{r} - \mathbf{r}'|}}{|\mathbf{r} - \mathbf{r}'|^2} dV(\mathbf{r}'), \quad (18)$$

where $\bar{a}_j$, defined as the "effective attenuation coefficient", is given by

$$\bar{a}_j(\mathbf{r}) = \frac{\int_0^{|\mathbf{r} - \mathbf{r}_j|} a(s'') ds''}{|\mathbf{r} - \mathbf{r}_j|}. \quad (19)$$

The effective attenuation coefficient is obtained by assuming that $|\mathbf{r} - \mathbf{r}'| \simeq |\mathbf{r} - \mathbf{r}_j|$ in evaluating the inner integral of Eq. (16). This approximation, justified by selecting small enough computational elements, simplifies computing the integral considerably. For a source-target configuration, the effective attenuation coefficient is calculated once, according to Eq. (19), and is then used as a constant in the triple integral (18).

Depending on the optical thickness of the source-target configuration, various simplifications of the expressions for $G_j(\mathbf{r})$ have been used.²⁰

Example

The example considered is that of an axisymmetric "ring" of methane reacting with air. The fuel ring consists initially of a region of fuel surrounded by air. Buoyancy is expected to play a dominant role since it is the major source of vorticity generation. The density change in time is also a source of volumetric expansion.

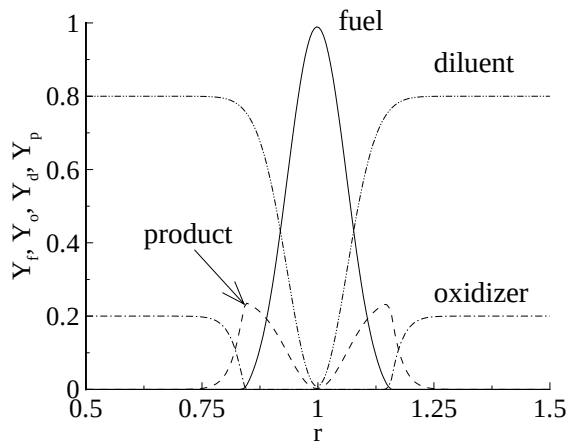


Fig. 2 Initial distribution of species' mass fractions presented in terms of radial profile at $z = 0$.

The vorticity and energy equations are coupled via the density and its gradients. The chemical reaction across the flame produces sharp density gradients and products of lower density.

The objectives are to establish numerical convergence and to investigate the impact of radiation on a reacting buoyant flow.

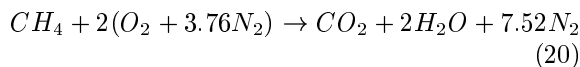
Assumptions

1. Unbounded axisymmetric domain.
2. Reynolds number is constant.
3. Prandtl number is unity.
4. Lewis number of all species is unity.
5. Chemical reaction is single step, irreversible and of infinite rate.
6. Gases are ideal, gray and non-scattering.

Parameters

The parameters are:

- Reynolds number: $Re = 50$.
- The mass stoichiometric ratio: $\phi = 4.0$.
- Heat of reaction: $Q_f \equiv h_f^0/c_{p0}T_0 = 80$, where h_f^0 is the heat of reaction of fuel per unit mass.
- The fuel is Methane.
- The chemical reaction is:



and the molar masses are: $M_f = 16$, $M_o = 32$, $M_d = 28$, and $M_p = 26.62$.

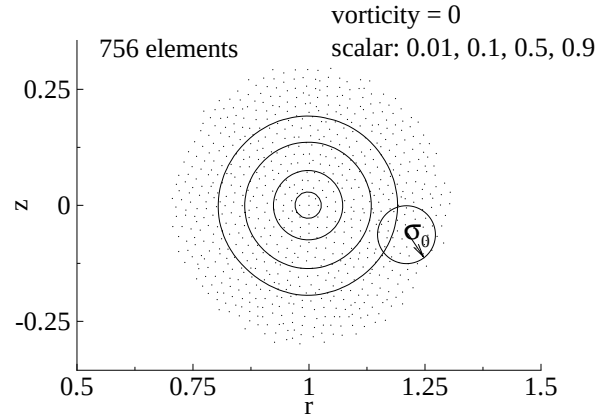


Fig. 3 Initial distribution of computational elements and contours of mixture fraction.

Initial conditions

The initial vorticity is zero. The initial temperature and species distributions, shown in Fig. 2 correspond to diffusion of a Dirac delta source located at $r_0 = 1$, $z_0 = 0$ over a time $\tau_0 = 0.002$, where $\tau \equiv t/Re$. The initial distribution, consisting of 756 of elements, shown in Fig. 3, was obtained by diffusing the point source using the smoothed redistribution method with $\Delta\tau_0 = 0.0002$, and the core radius was increased from $\sigma = 0$ to $\sigma_0 = \sqrt{20\Delta\tau_0}$. Contours of the mixture fraction are also shown in the figure. The mixture fraction varies from zero on the air side to one on the fuel side. The boundary conditions are:

on the fuel side: $Y_f = 1$, $Y_o = 0$, $Y_p = 0$, $Y_d = 0$, $T = 1$.

on the air side: $Y_f = 0$, $Y_o = 0.2$, $Y_p = 0$, $Y_d = 0.8$, $T = 1$.

Boundary conditions

- At infinity:

$$\omega_{\mathbf{r} \rightarrow \infty} = 0 \quad \text{and} \quad \mathbf{s}_{\mathbf{r} \rightarrow \infty} = \mathbf{s}_0. \quad (21)$$

- At axis of symmetry:

$$\omega_{r=0} = 0 \quad \text{and} \quad \left(\frac{\partial s}{\partial r} \right)_{r=0} = 0. \quad (22)$$

Numerical parameters

The computations were conducted with the following numerical parameters: $h = \sqrt{8\Delta\tau}$, $\Delta\tau = 0.0005$, 0.001, 0.002, and 0.005, and the final time is $\tau_f = 0.2$, where h is the initial elements spacing. The core expansion fraction of the diffusion step and the upper bound of the core radius were chosen to be $x = 0.1$ and $\sigma_{max} = 4h$. Unless otherwise specified, results discussed correspond to $\Delta\tau = 0.0005$.

Results and discussion

The case of the reacting fuel ring without radiation is discussed first. The role of radiation is discussed next.

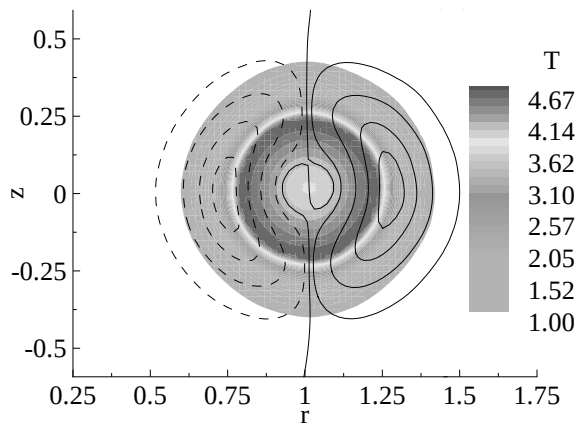


Fig. 4 Vorticity distribution at $\tau = 0.005$. Dashed contours: -1.4, -1, -0.5 and -0.1. Solid contours: 0, 0.1, 0.5, 1 and 1.4. Temperature distribution is also shown.

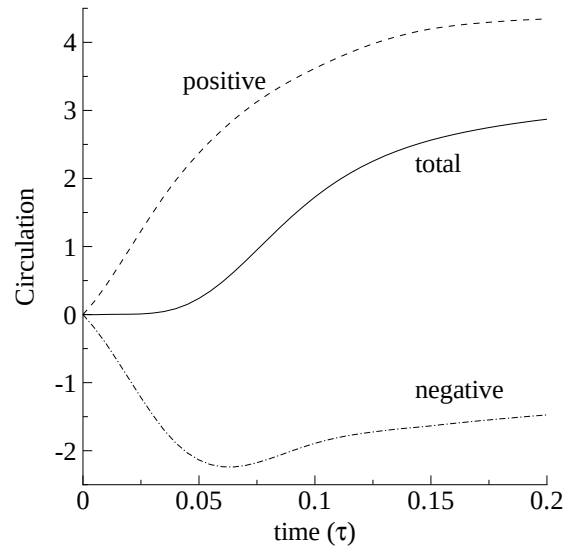


Fig. 6 Total positive, negative, and net circulation as function of time.

Vorticity

In the early stages, the temperature distribution resembles that of a two dimensional planar problem; it is symmetrical around the r and z axes passing through the ring center. The corresponding vorticity distribution, generated due to the interaction between temperature gradients and gravity, is also symmetric. This is confirmed in Fig. 4, where vorticity contours are overlayed on top of the temperature distribution. Note that positive vorticity is associated with negative temperature gradient and vice versa. These two symmetrical regions of opposite vorticity balance each other so that the flow is dominated by diffusion and convection plays a minor role. However, as the vorticity spreads, it starts dissipating across the z axis. Thus, diffusion across the axis causes the negative vorticity to decrease in absolute value, so that the net overall circulation is positive causing the overall structure to rotate clockwise. This is confirmed in Fig. 5 where the vorticity contours are plotted at four times $\tau = 0.05, 0.1, 0.15$ and 0.2 . The symmetrical character of the vorticity at initial times can still be seen at $\tau = 0.05$. At later times, the structure rises and rotates clockwise. Assuming that diffusion is the dominant process initially, and knowing that vorticity is generated at the outer edge where products form, then according to the solution of the diffusion equation, it takes $\tau \sim 0.06$ for the negative vorticity region to reach the z axis and start decaying. Indeed, this is the case as can be seen in Fig. 6, where the positive, negative, and total circulation (positive + negative) are plotted. The net total circulation stays close to zero until $\tau \sim 0.05$ after which it starts increasing.

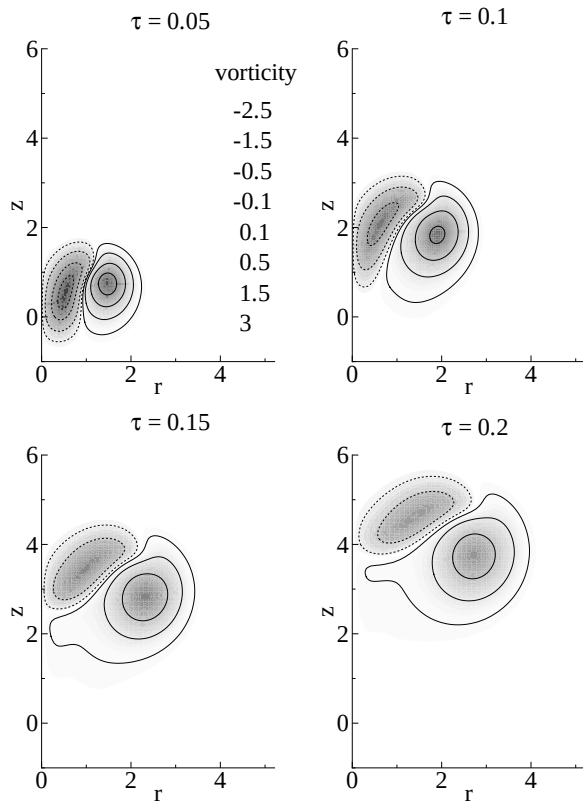


Fig. 5 Vorticity distribution at $\tau = 0.05, 0.1, 0.15$, and 0.2 . Dashed contours: -2.5, -1.5, -0.5 and -0.1. Solid contours: 0.1, 0.5, 1.5 and 3.

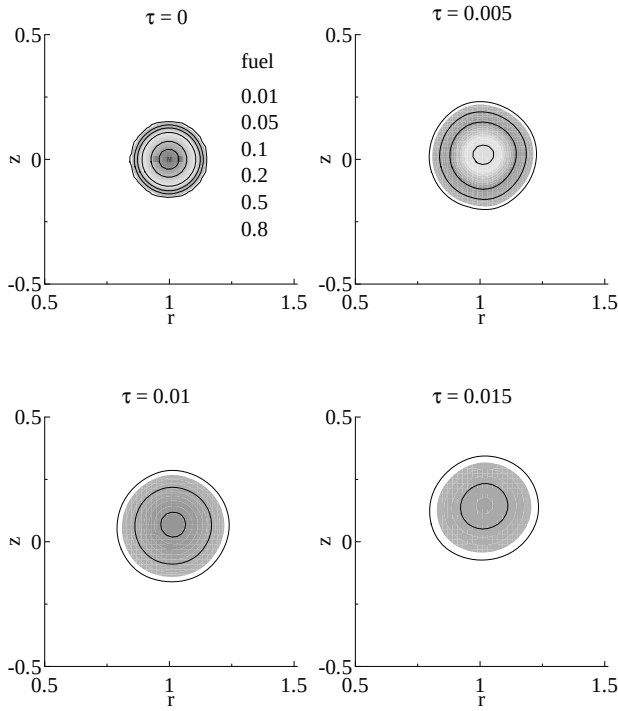


Fig. 7 Fuel distribution at $\tau = 0.05, 0.1, 0.15$, and 0.2 . Contour values: $0.01, 0.05, 0.1, 0.2, 0.5$, and 0.8

Scalars

The fuel is consumed at the rate it diffuses to the reaction zone. At the early stages, the solution of the scalar diffusion equation is given by the Green's function

$$s(\mathbf{r}; \mathbf{r}_0, t) = \frac{2\pi E}{\pi^{3/2}\sigma^3} e^{-\frac{r^2 + r_0^2 + (z - z_0)^2}{\sigma^2}} I_0\left(\frac{2rr_0}{\sigma^2}\right), \quad (23)$$

where E is energy and I_0 is the modified Bessel function of the first kind and order zero. Starting with the initial scalar distribution, the scalar at the center of the ring, i.e. at $r = 1, z = 0$ decreases to 10 % of its initial value at $\tau = 0.015$. This is confirmed in Fig. 7 where at $\tau = 0.015$, the fuel at the center of the ring is about 8 % of its initial value. The fuel mass fraction distribution presented in the figure shows that most of fuel consumption occurs at the early stages. This is reflected by the symmetry in the distribution that resembles the solution of the diffusion equation. The fact that the numerical value is less than that obtained by pure diffusion may be attributed to the combined role of volumetric expansion and convection.

The temperature distribution at times $\tau = 0.05, 0.1, 0.15$, and 0.2 are shown in Fig. 8. As mentioned in the previous section, most of the fuel has been consumed before $\sim \tau = 0.02$, and consequently no chemical reaction occurs afterwards. The temperature distribution is then governed by diffusion and convection, the later due to vorticity and volumetric expansion. Contrary to the early stages where temperature is largest in the

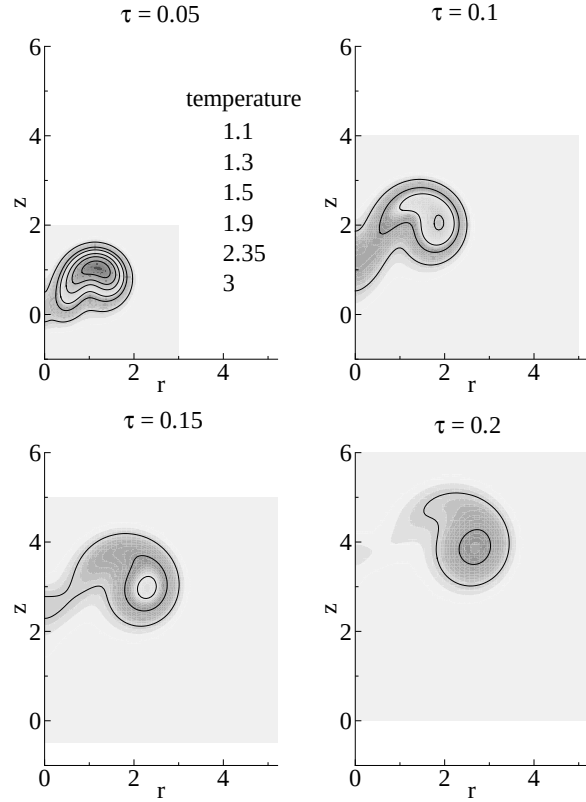


Fig. 8 Temperature distribution at $\tau = 0.05, 0.1, 0.15$, and 0.2 . Contour values: $1.1, 1.3, 1.5, 1.9, 2.35$ and 3 .

reaction zone as seen in Fig. 4, large temperatures at later stages are associated with regions of high products concentration, as seen in Fig. 8. Regions of large vorticity are associated with high temperature gradients as seen in vorticity and temperature distributions shown in Figs. 5 and 8 respectively. Finally, the impact of volumetric expansion is manifested in larger volume of hot gas with a mean center moving away from the axis.

Convergence

To investigate numerical convergence the following time steps were used: $\Delta\tau = 0.0005, 0.001, 0.002$ and 0.005 . The results are compared in terms of vorticity and scalar contours at the last time step, i.e. at $\tau = 0.2$. The vorticity contours presented in Fig. 9 show that the numerical solution converges in a nonlinear fashion. This is confirmed in Fig. 10 where the mixture fraction contours are shown.

Using the solution with $\Delta\tau = 0.0005$ as a reference, the departure of the solutions for $\Delta\tau = 0.001, 0.002$ and 0.005 from this reference is measured in terms of an L-2 norm of the total error evaluated on a uniform mesh. The errors for both vorticity and scalar are presented in Fig. 11 as a function of $\Delta\tau$. The error behaves as $\Delta\tau^m$, where $m = 1.52$ for vorticity and 1.27 for scalar.

The number of elements as a function of time is

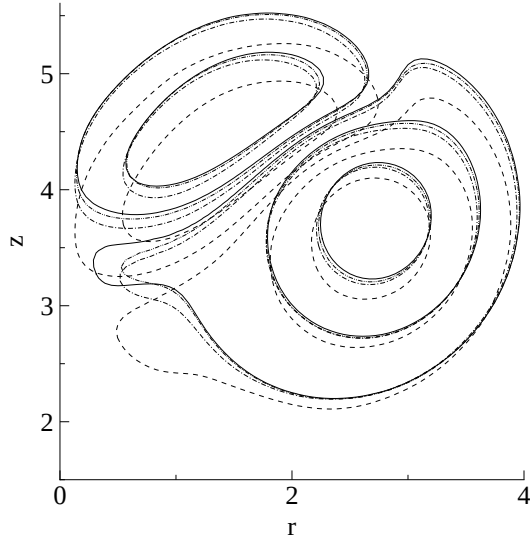


Fig. 9 Vorticity distribution at $\tau = 0.2$. Contour values: -2.5, -1.5, -0.5, -0.1, 0.1, 0.5, 1.5, and 3. Solid line: $\Delta\tau = 0.0005$. Dash-dot-dot line: $\Delta\tau = 0.001$. Dash-dot line: $\Delta\tau = 0.002$. Dashed line: $\Delta\tau = 0.005$.

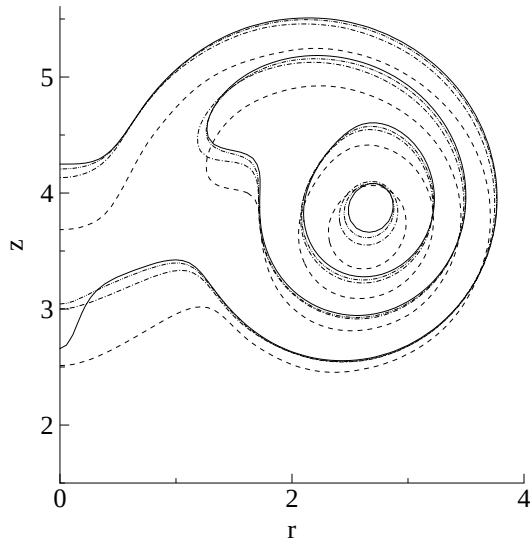


Fig. 10 Mixture fraction distribution at $\tau = 0.2$. Contour values: 0.00025, 0.001, 0.0025 and 0.0045. Solid line: $\Delta\tau = 0.0005$. Dash-dot-dot line: $\Delta\tau = 0.001$. Dash-dot line: $\Delta\tau = 0.002$. Dashed line: $\Delta\tau = 0.005$.

presented in Fig. 12 for $\Delta\tau = 0.0005$. The increase in number of elements is linear. Further, it was observed that it decreases linearly with $\Delta\tau$. For example, the number of elements at $\tau = 0.2$ is 13368, 7764, and 4173 for $\Delta\tau = 0.0005$, 0.001 and 0.002 respectively.

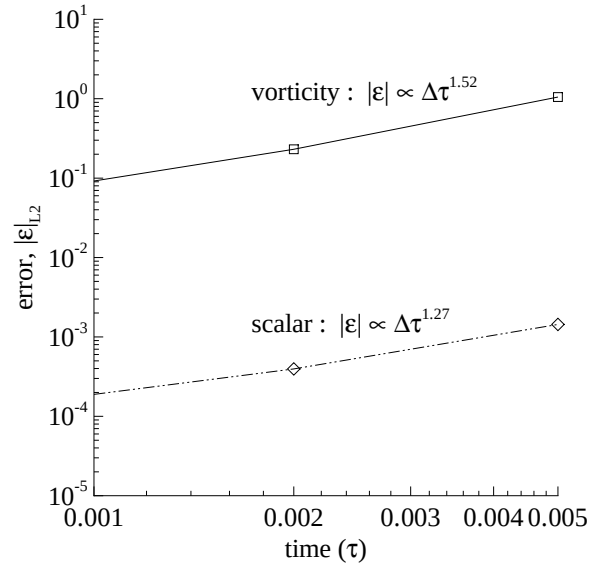


Fig. 11 L-2 norm of error in vorticity and mixture fraction as a function of time step.

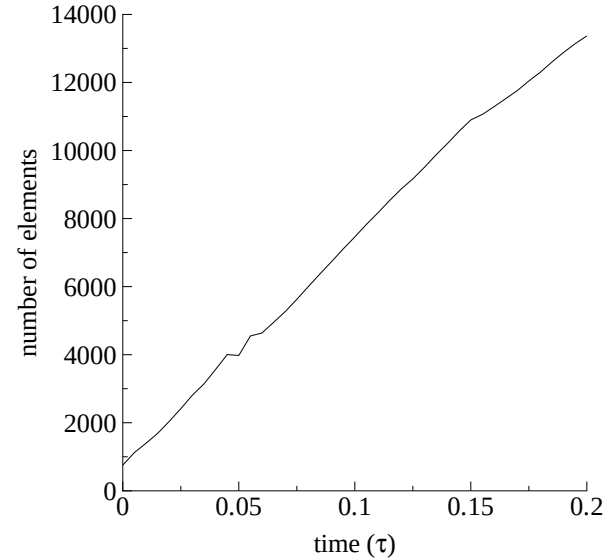


Fig. 12 Number of elements versus time for $\Delta\tau = 0.0005$.

The role of radiation

In order to investigate the impact of radiation, the same example has been simulated, in which the radiation source term in the energy equation is determined according to the discrete source method described in the numerical methodology section. The medium is assumed to be gray, i.e. wavelength independent. The absorption coefficient is determined according to²⁸

$$a = \sum a_{p,i}(T)p_i \quad (24)$$

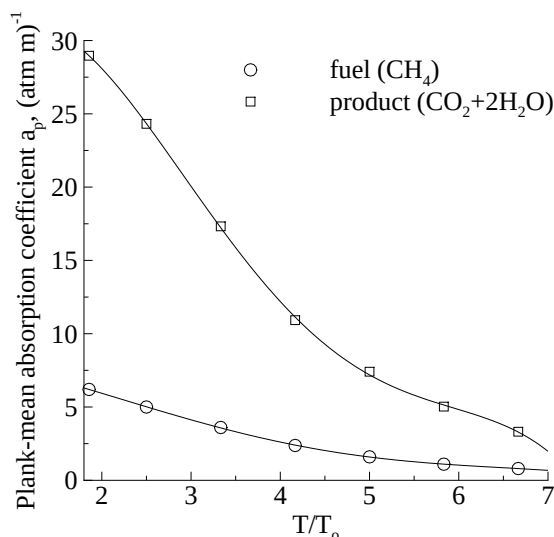


Fig. 13 Dependence of the Plank-mean absorption coefficient on temperature.

where i corresponds to chemical species, and $a_{p,i}$ is the Plank-mean absorption coefficient of species i , and p_i is the partial pressure of the species. The temperature dependence of a_p for the various species, obtained using narrow band averaged values,³⁰ is shown in Fig. 13. The absorption coefficient of air is negligible for the range of temperature of interest.

The vorticity distribution, shown in Fig. 14, resembles that of the non-radiating ring except that slightly lower values of vorticity are noticed. This is attributed to smaller vorticity generation associated with weaker temperature gradients. The temperature distribution is presented in Fig. 15. Radiation contributes to the temperature field in two ways: first it results in thermal energy loss to the surrounding and second it redistributes the thermal energy in an equilibrating manner, resulting in smaller temperature gradients.

A comparison of the time variation of the total circulation with and without radiation is shown in Fig. 16. Radiation results in reduction in the absolute value of the total negative and positive circulation generation with a reduction in the net overall circulation. This may be thought of as a measure of the role of radiation in reducing vorticity generation via lower temperature gradients.

The total energy loss due to radiation is plotted as a function of time in Fig. 17. During the stage where $\tau < 0.05$, the rate of energy loss is large as can be seen from the steep rate of energy lost. Later, for $\tau > 0.05$, energy loss approaches a steady state value reflecting the decaying role of radiation. This can be explained in terms of the corresponding fuel consumption. The maximum fuel mass fraction in the field is plotted as a function of time in the same figure. It can be noticed that the rate of energy loss due to radiation starts approaching zero by the time all the fuel has been

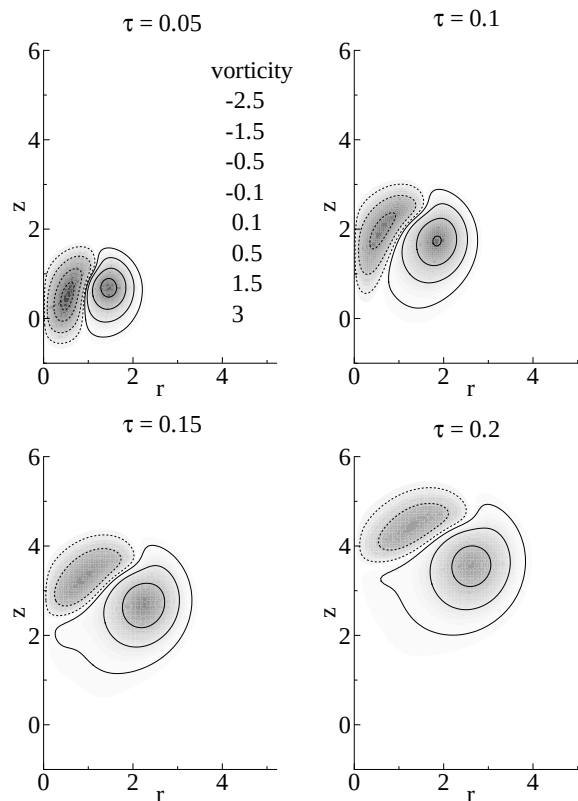


Fig. 14 Vorticity distribution for the radiating case at $\tau = 0.05, 0.1, 0.15$, and 0.2 . Dashed contours: $-2.5, -1.5, -0.5$ and -0.1 . Solid contours: $0.1, 0.5, 1.5$ and 3 .

consumed, represented by the dashed line in the figure. The shift in the radiation loss profile is due to the high thermal energy of the products of reaction right after the complete consumption of the fuel.

Conclusion

Grid-free Lagrangian computations of a reacting fuel ring with and without radiation have been presented. Radiation modeling is based on the discrete source method; a recent development in vortex methods. Radiation plays the roles of redistribution of thermal energy and losing part of it to the surrounding. Initially, the vorticity generated by buoyancy is symmetrical. Later diffusion causes the negative vorticity close to the axis to decay in absolute value resulting in rise and clockwise rotation of the structure. Radiation, via reducing temperature gradients, results in smaller generation (in absolute value) of positive, negative, and net overall circulation. Future works involves investigation of the impact of higher Reynolds number and heat release.

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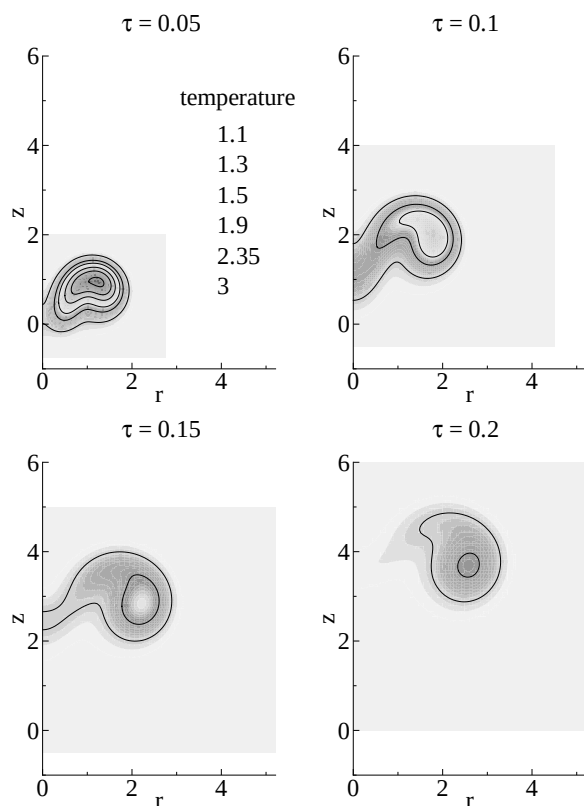


Fig. 15 Temperature distribution for the radiating case at $\tau = 0.05, 0.1, 0.15$, and 0.2 . Contour values: 1.1, 1.3, 1.5, 1.9, 2.35, and 3.

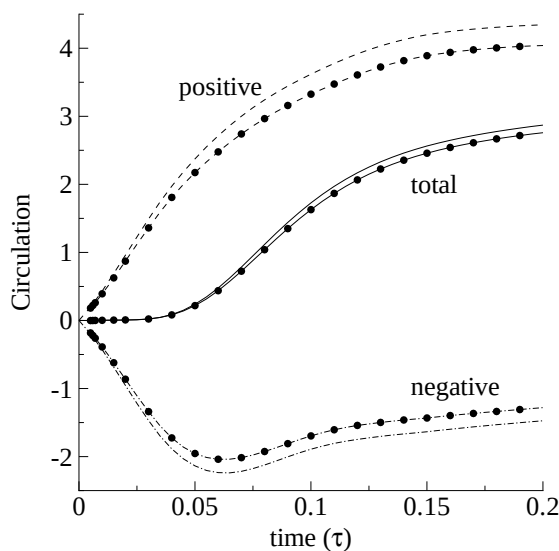


Fig. 16 Total positive, negative, and net circulation as function of time with and without radiation. Symbols denote the case with radiation.

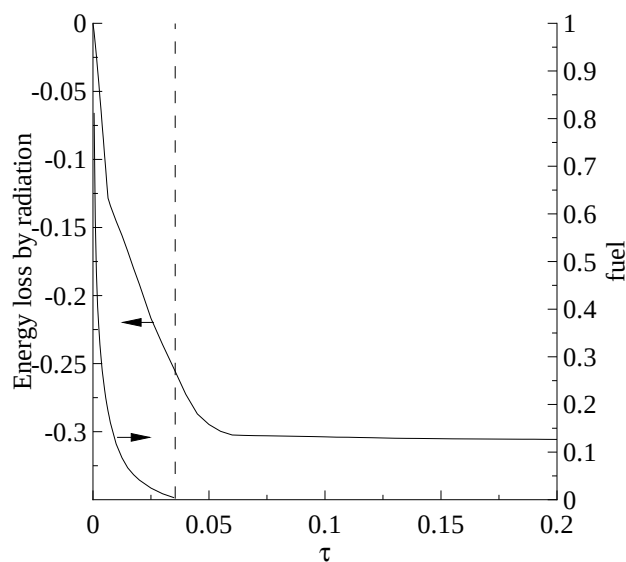


Fig. 17 Ratio of thermal energy loss to energy generated by chemical reaction as a function of time. The maximum fuel mass fraction is also presented.

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